

05 - A2A Engine (Application-to-Application)

Overview

The A2A Engine enables direct machine-to-machine electronic return submission to the IRS Modernized e-File system using the SOAP-based A2A protocol, eliminating manual intermediary steps.

A2A Protocol Stack

PractitionerHub A2A Engine

- > SOAP Envelope (XML)
 - > HTTPS/mTLS (PFX Certificate for EFIN 748335)
 - > IRS MeF A2A Gateway (la.www4.irs.gov/a2a/)
 - > IRS MeF Processing System
-

Supported Submission Types

Form	Description	Tax Year Support
1040	Individual Income Tax	Current + 2 prior
1040-SR	Senior Individual	Current + 2 prior
1040-NR	Nonresident Alien	Current + 2 prior
1120	C-Corporation	Current + 2 prior
1120-S	S-Corporation	Current + 2 prior
1065	Partnership	Current + 2 prior
990	Exempt Organization	Current + 2 prior
941	Quarterly Payroll	Current year
4868	Extension of Time	Current year
2350	Extension (Abroad)	Current year

SOAP Envelope Structure

```
<soapenv:Envelope
  xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:irs="http://www.irs.gov/a2a">
  <soapenv:Header>
    <irs:EFINHeader>
      <irs:EFIN>748335</irs:EFIN>
      <irs:SubmissionType>Individual</irs:SubmissionType>
      <irs:TaxYear>2025</irs:TaxYear>
    </irs:EFINHeader>
  </soapenv:Header>
  <soapenv:Body>
    <irs:SubmissionManifest>
      <irs:SubmissionId>[UUID]</irs:SubmissionId>
      <irs:ReturnType>1040</irs:ReturnType>
      <irs:TaxPeriodBeginDate>2025-01-01</irs:TaxPeriodBeginDate>
      <irs:TaxPeriodEndDate>2025-12-31</irs:TaxPeriodEndDate>
    </irs:SubmissionManifest>
    <!-- Return data attachment (Base64 encoded ZIP) -->
  </soapenv:Body>
</soapenv:Envelope>
```

Acknowledgment Codes

Code	Meaning	Next Action
A	Accepted	Notify client and archive
R	Rejected	Parse error codes, correct, resubmit
P	Pending	Poll again in 90 seconds
I	Incomplete	Add missing data and resubmit
D	Duplicate	Do not resubmit, investigate

Retry Configuration

```
a2a_engine:
  endpoint: "https://la.www4.irs.gov/a2a/"
  efin: "748335"
  cert_path: "/certs/mef-a2a-748335.pfx"
  batch_size_max: 100
  retry_attempts: 3
  retry_backoff: [30, 60, 120]
  ack_poll_interval_seconds: 90
  ack_timeout_hours: 48
  soap_version: "1.1"
  compression: "ZIP"
  encryption: "AES-256"
```

Submission Audit Log Fields

Field	Type	Description
submission_id	UUID	Unique per submission
efin	String	748335
form_type	String	1040, 1120, and others
tax_year	Integer	Example: 2025
tin_masked	String	XXX-XX-1234
transmitted_at	ISO 8601	UTC timestamp
ack_code	Enum	A/R/P/I/D
ack_received_at	ISO 8601	UTC timestamp
preparer_id	String	Internal preparer reference
submission_file_hash	SHA-256	File integrity check